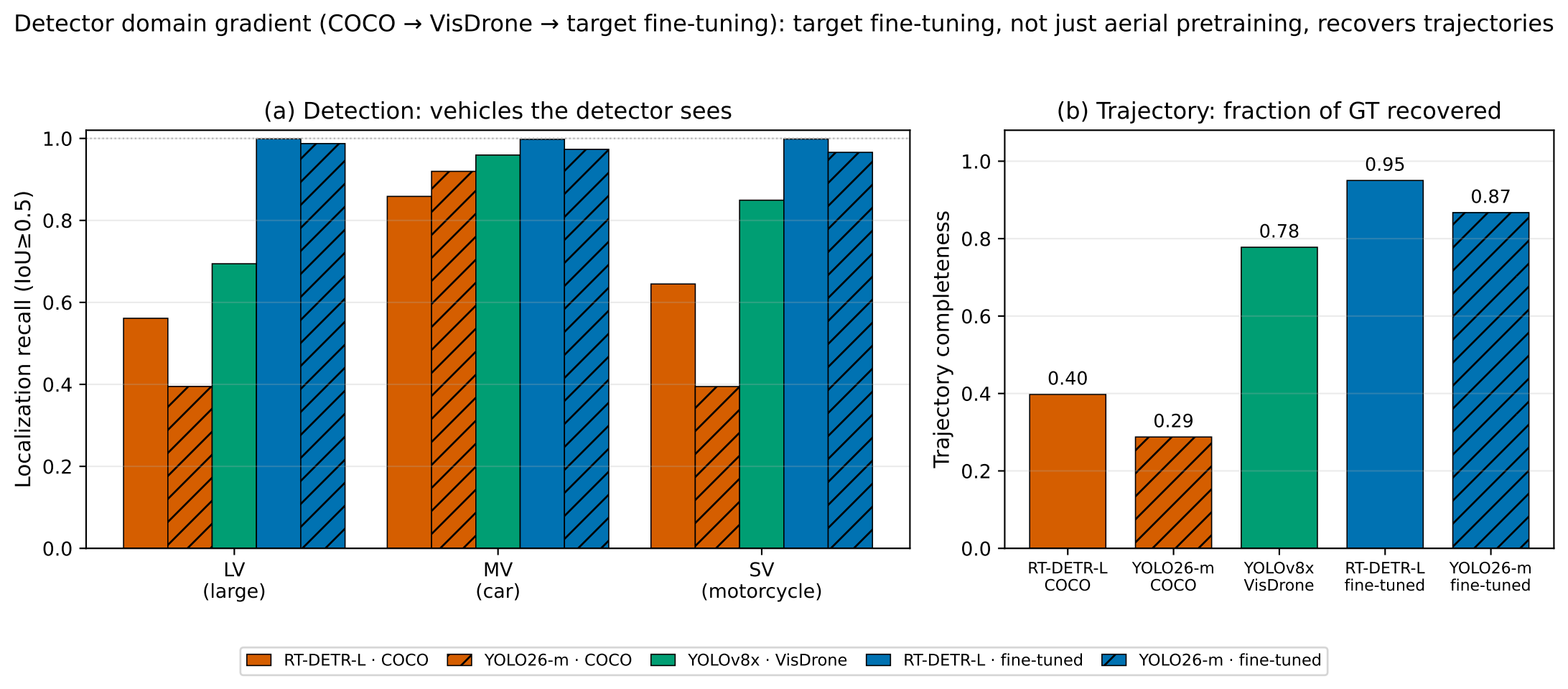
Detector domain gradient (COCO → VisDrone → target fine-tuning): target fine-tuning, not just aerial pretraining, recovers trajectories
(a) Detection: vehicles the detector sees
Localization recall (IoU≥0.5)
0.0
0.2
0.4
0.6
0.8
1.0
LV
(large)
MV
(car)
SV
(motorcycle)
(b) Trajectory: fraction of GT recovered
Trajectory completeness
0.40
0.29
0.78
0.95
0.87
RT-DETR-L
COCO
YOLO26-m
COCO
YOLOv8x
VisDrone
RT-DETR-L
fine-tuned
YOLO26-m
fine-tuned
RT-DETR-L · COCO
YOLO26-m · COCO
YOLOv8x · VisDrone
RT-DETR-L · fine-tuned
YOLO26-m · fine-tuned